

Statistical Modeling

DATA.STAT.740

Lecture 5

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Example: Normal distribution is from the exponential family of distributions

$$\mathbf{y}_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\begin{aligned} f(y_i \mid \mu_i, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(y_i - \mu_i)^2}{\sigma^2}\right) \\ &= \exp\left(-\frac{1}{2} \frac{y_i^2 - 2y_i\mu_i + \mu_i^2}{\sigma^2} + \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)\right) \\ &= \exp\left(\frac{y_i\mu_i - \frac{1}{2}\mu_i^2}{\sigma^2} - \frac{y_i^2}{2\sigma^2} + \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)\right) \\ &= \exp\left(\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right) \end{aligned}$$

where

$$\theta_i = \mu_i, \quad \phi = \sigma^2, \quad a(\phi) = \phi, \quad b(\theta_i) = \frac{1}{2}\theta_i^2, \quad c(y_i, \phi) = -\frac{y_i^2}{2\phi} + \log\left(\frac{1}{\sqrt{2\pi\phi}}\right)$$

How about the mean and the variance? $\mu_i = \mathbb{E}(\mathbf{y}_i \mid x_i) = b'(\theta_i), \text{Var}(\mathbf{y}_i \mid x_i) = b''(\theta_i)a(\phi)$

Example: Bernoulli distribution is from the exponential family of distributions

$y_i \sim \text{Bernoulli}(\mu_i)$

$$\begin{aligned} f(y_i | \mu_i) &= \mu_i^{y_i} (1 - \mu_i)^{(1-y_i)} \\ &= \exp \left(\log \left(\mu_i^{y_i} (1 - \mu_i)^{(1-y_i)} \right) \right) \\ &= \exp (y_i \log (\mu_i) + (1 - y_i) \log (1 - \mu_i)) \\ &= \exp \left(y_i \log \left(\frac{\mu_i}{1 - \mu_i} \right) + \log(1 - \mu_i) \right) \end{aligned}$$

Let's define

$$\theta_i = \log \left(\frac{\mu_i}{1 - \mu_i} \right) \Leftrightarrow \mu_i = \frac{\exp(\theta)}{1 + \exp(\theta)}$$

Now

$$\log(1 - \mu_i) = \log \mu_i - \theta_i = \log \left(\frac{\exp(\theta_i)}{1 + \exp(\theta_i)} \right) - \log(\exp(\theta_i)) = \log \left(\frac{1}{1 + \exp(\theta_i)} \right)$$

Example: Bernoulli distribution is from the exponential family of distributions

$$\mathbf{y}_i \sim \text{Bernoulli}(\mu_i)$$

$$f(y_i | \mu_i) = \exp \left(y_i \log \left(\frac{\mu_i}{1 - \mu_i} \right) - \log (1 + \exp(\theta_i)) \right)$$

Now

$$\theta_i = \log \left(\frac{\mu_i}{1 - \mu_i} \right), \quad b(\theta_i) = \log (1 + \exp(\theta_i)), \quad c(y_i) = 0$$

If the distribution is a single-parameter distribution such as Bernoulli and Poisson distribution, then there is no dispersion parameter and the format is

$$f(y_i | \theta_i, \phi) = \exp (y_i \theta_i - b(\theta_i) + c(y_i))$$

How about the mean and the variance?

$$\mu_i = \mathbb{E}(\mathbf{y}_i | x_i) = b'(\theta_i), \quad \text{Var}(\mathbf{y}_i | x_i) = b''(\theta_i)a(\phi)$$

Likelihood in exponential family of distributions

Let there be i.i.d observations $y_1, \dots, y_N$

Likelihood and log-likelihood are now

$$\prod_{i=1}^N L(\theta_i, \phi) = \prod_{i=1}^N \exp \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$$
$$\sum_{i=1}^N l(\theta_i, \phi) = \sum_{i=1}^N \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$$

Maximum likelihood

We want to find the parameters β that maximize the likelihood

$$\sum_{i=1}^N l(\theta_i, \phi) = \sum_{i=1}^N \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$$

by taking the partial derivate with respect to β

Now $\beta \rightarrow \eta_i = x_i^T \beta \rightarrow \mu_i = g^{-1}(\eta_i) \rightarrow \theta_i$ with $\mu_i = b'(\theta_i)$

Finding the maximum likelihood

$$\frac{\partial}{\partial \beta} \sum_{i=1}^N l(\theta_i, \phi) = \sum_{i=1}^N \frac{\partial}{\partial \beta} \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$$

Use the chain rule $\frac{\partial l(\theta_i, \phi)}{\partial \beta} = \frac{\partial l(\theta_i, \phi)}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta}$

Calculate terms separately $\frac{\partial l(\theta_i, \phi)}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{a(\phi)}$

$$\begin{aligned} \mu_i = b'(\theta_i) &\Rightarrow \frac{\partial \mu_i}{\partial \theta_i} = b''(\theta_i) \Rightarrow \frac{\partial \theta_i}{\partial \mu_i} = \frac{1}{b''(\theta_i)} \\ \frac{\partial \mu_i}{\partial \eta_i} &= \frac{\partial g^{-1}(\eta_i)}{\partial \eta_i} = (g^{-1})'(\eta_i) \\ \frac{\partial \eta_i}{\partial \beta} &= \frac{\partial x_i^T \beta}{\partial \beta} = x_i \end{aligned}$$

Score equations

Put the components into the chain rule $\frac{\partial l(\theta_i, \phi)}{\partial \beta} = \frac{\partial l(\theta_i, \phi)}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta}$

and formulate the set of equations

$$\sum_{i=1}^N \frac{y_i - b'(\theta_i)}{a(\phi)} \cdot \frac{1}{b''(\theta_i)} \cdot (g^{-1})'(\eta_i) \cdot x_i = 0$$

if we use the equalities $\frac{1}{a(\phi)b''(\theta_i)} = \frac{1}{\text{Var}(\mathbf{y}_i)}$, $\mu_i = b'(\theta_i)$ and $\frac{\partial \mu_i}{\partial \eta_i} = (g^{-1})'(\eta_i)$

We can formulate the **score equations**

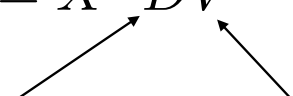
$$\sum_{i=1}^N \frac{y_i - \mu_i}{\text{Var}(\mathbf{y}_i)} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot x_i = 0$$

Score equations

The general formulation for GLMs is that we solve the score equations for β

$$\sum_{i=1}^N \frac{y_i - \mu_i}{\text{Var}(\mathbf{y}_i)} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot x_i = 0$$

where the dependence of β is through $\mu_i = g^{-1}(x_i^T \beta)$

$$\mathbf{u}_\beta = \frac{\partial l(\beta \mid \mathbf{y})}{\partial \beta} = X^T D V^{-1} (\mathbf{y} - \mu)$$

$$D = \begin{bmatrix} \frac{\partial \mu_1}{\partial \eta_1} & 0 & \cdots & 0 \\ 0 & \frac{\partial \mu_2}{\partial \eta_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\partial \mu_N}{\partial \eta_N} \end{bmatrix} \quad V = \begin{bmatrix} \text{Var}(\mathbf{y}_1) & 0 & \cdots & 0 \\ 0 & \text{Var}(\mathbf{y}_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \text{Var}(\mathbf{y}_N) \end{bmatrix}$$

Maximum likelihood estimator

The general formulation for GLMs is that we solve the score equations for β

$$\sum_{i=1}^N \frac{y_i - \mu_i}{\text{Var}(\mathbf{y}_i)} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot x_i = 0$$

where the dependence of β is through $\mu_i = g^{-1}(x_i^T \beta)$

$$\mathbf{u}_\beta = \frac{\partial l(\beta \mid \mathbf{y})}{\partial \beta} = X^T D V^{-1}(\mathbf{y} - \mu)$$

Estimator $\hat{\beta}$ for which $\mathbf{u}_{\hat{\beta}} = X^T \hat{D} \hat{V}^{-1}(\mathbf{y} - \hat{\mu}) = 0$, is called **maximum likelihood estimator** of β

Newton-Rhapson method

We can't find $\hat{\beta}$ analytically

$$\mathbf{u}_{\hat{\beta}} = X^T \hat{D} \hat{V}^{-1} (\mathbf{y} - \hat{\mu}) = 0$$

but we need to find it using e.g. iterative optimization algorithms

Newton-Rhapson method provides successively better approximations to zeroes of a real-valued function

It is based on 2nd degree Taylor approximation of the likelihood around the current estimate β_t (we set the initial value β_0)

$$l(\beta) \approx l(\beta_t) + \mathbf{u}_t^T (\beta - \beta_t) + \frac{1}{2} (\beta - \beta_t)^T H_t (\beta - \beta_t)$$

Newton-Rhapson and Fisher scoring

Approximated likelihood

$$l(\beta) \approx l(\beta_t) + \mathbf{u}_t^T (\beta - \beta_t) + \frac{1}{2} (\beta - \beta_t)^T H_t (\beta - \beta_t)$$

$\nwarrow$
 Hessian $H_t = \begin{bmatrix} \frac{\partial^2 l(\beta)}{\partial \beta_0 \partial \beta_0} & \frac{\partial^2 l(\beta)}{\partial \beta_0 \partial \beta_1} & \cdots & \frac{\partial^2 l(\beta)}{\partial \beta_0 \partial \beta_p} \\ \frac{\partial^2 l(\beta)}{\partial \beta_1 \partial \beta_0} & \frac{\partial^2 l(\beta)}{\partial \beta_1 \partial \beta_1} & \cdots & \frac{\partial^2 l(\beta)}{\partial \beta_1 \partial \beta_p} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial^2 l(\beta)}{\partial \beta_p \partial \beta_0} & \frac{\partial^2 l(\beta)}{\partial \beta_p \partial \beta_1} & \cdots & \frac{\partial^2 l(\beta)}{\partial \beta_p \partial \beta_p} \end{bmatrix}_{\beta=\beta_t}$

Calculate the partial derivatives and set to 0

$$\frac{\partial l(\beta)}{\partial \beta} \approx \mathbf{u}_t + H_t (\beta - \beta_t) = 0$$

Solve for β and set it as the next estimate β_{t+1}

$$\beta_{t+1} = \beta_t - H_t^{-1} \mathbf{u}_t$$

Fisher scoring is Newton-Rhapson where $F_t = -E(H_t)$

Fisher scoring

Fisher scoring is similar to Newton-Rhapson $\beta_{t+1} = \beta_t - H_t^{-1} \mathbf{u}_t$ where we replace H_t with Fisher information matrix

$$F_t = -\mathbb{E}(H_t)$$

Therefore Fisher scoring has an update

$$\beta_{t+1} = \beta_t + F_t^{-1} \mathbf{u}_t$$

Newton-Rhapson and Fisher scoring both continue until $|\beta_{t+1} - \beta_t|$ is sufficiently close to zero - and approximate

$$\hat{\beta} \approx \beta_{t+1}$$

Weighted least squares estimation

Let $\hat{\beta}$ be the maximum likelihood **estimator** for the GLM $g(\mu) = \eta = X\beta$

From the Fisher scoring method we assume that the difference between the true parameter and our the ML estimator is

$$\hat{\beta} - \beta = F_{\hat{\beta}}^{-1} \mathbf{u}_{\hat{\beta}}$$

We assume that $F_{\hat{\beta}} \approx F_{\beta}$ and $\mathbf{u}_{\hat{\beta}} \approx \mathbf{u}_{\beta}$

Therefore

$$\mathbb{E}(\hat{\beta} - \beta) = \mathbb{E}(F_{\beta}^{-1} \mathbf{u}_{\hat{\beta}}) = F_{\beta}^{-1} \mathbb{E}(\mathbf{u}_{\hat{\beta}}) = F_{\beta}^{-1} \cdot 0 = 0$$

$$\begin{aligned} \text{Cov}(\hat{\beta}) &= \mathbb{E} \left(\left(\hat{\beta} - \beta \right) \left(\hat{\beta} - \beta \right)^T \right) = \mathbb{E} \left(F_{\beta}^{-1} \mathbf{u}_{\hat{\beta}} \mathbf{u}_{\hat{\beta}}^T F_{\beta}^{-1} \right) \\ &= F_{\beta}^{-1} \mathbb{E} \left(\mathbf{u}_{\hat{\beta}} \mathbf{u}_{\hat{\beta}}^T \right) F_{\beta}^{-1} = F_{\beta}^{-1} F_{\beta} F_{\beta}^{-1} = F_{\beta}^{-1} \end{aligned}$$

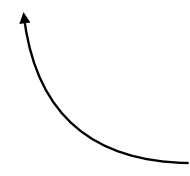
Weighted least squares estimation

Let $\hat{\beta}$ be the maximum likelihood **estimator** for the GLM $g(\mu) = \eta = X\beta$

Asymptotically we can approximate $\hat{\beta} \sim \text{Normal}(\beta, F_{\beta}^{-1})$

where

$$\begin{aligned} F_{\beta} &= \text{E} \left(X^T D V^{-1} (\mathbf{y} - \mu) (\mathbf{y} - \mu)^T V^{-1} D X \right) \\ &= X^T D V^{-1} \text{E} \left((\mathbf{y} - \mu) (\mathbf{y} - \mu)^T V^{-1} D X \right) \\ &= X^T D V^{-1} V V^{-1} D X = X^T W X \end{aligned}$$



$$W = D V^{-1} D = \begin{bmatrix} \frac{\frac{\partial \mu_1}{\partial \eta_1}}{\text{Var}(\mathbf{y}_1)} & 0 & \cdots & 0 \\ 0 & \frac{\frac{\partial \mu_2}{\partial \eta_2}}{\text{Var}(\mathbf{y}_2)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\frac{\partial \mu_N}{\partial \eta_N}}{\text{Var}(\mathbf{y}_N)} \end{bmatrix}$$

Weighted least squares estimation

The Fisher scoring process can be written as

$$\begin{aligned}\beta_{t+1} &= \beta_t + F_t^{-1} \mathbf{u}_t \\ F_t \beta_{t+1} &= F_t \beta_t + \mathbf{u}_t\end{aligned}$$

and setting $F_t = X^T W_t X$, $\mathbf{u}_t = X^T D_t V^{-1}(\mathbf{y} - \mu_t) = X^T W_t D_t^{-1}(\mathbf{y} - \mu_t)$
we get

$$\begin{aligned}X^T W_t X \beta_{t+1} &= X^T W_t (X \beta_t + D_t^{-1}(\mathbf{y} - \mu_t)) \\ X^T W_t X \beta_{t+1} &= X^T W_t \mathbf{z}_t\end{aligned}$$

where $\mathbf{z}_t = X \beta_t + D_t^{-1}(\mathbf{y} - \mu_t)$ is **adjusted response variable**

and $\beta_{t+1} = (X^T W_t X)^{-1} X^T W_t \mathbf{z}_t$ is **the weighted least squares estimate**

Confidence intervals

There are multiple ways of estimating confidence intervals for the GLM coefficients estimates $\hat{\beta}_i$

Most common approaches are

1. Wald confidence intervals

Routine work and large samples

2. Profile likelihood intervals

When accuracy is more important and normality assumption is doubtful

3. Bootstrap intervals (not talked about today)

When theory-based approximations are doubtful

Wald confidence intervals

Based on asymptotic approximation

$$\hat{\beta} \sim \text{Normal}(\beta, F_{\beta}^{-1})$$

For β_j the standard error of the ML estimator is $\widehat{\text{SE}}(\hat{\beta}_j) = \sqrt{[F_{\beta}^{-1}]_{jj}}$

$$100(1 - \alpha)\% - \text{CI: } \hat{\beta}_j \pm z_{1-\alpha/2} \widehat{\text{SE}}(\hat{\beta}_j)$$

Fast and often reported by software

Profile likelihood intervals

For each parameter β_j , $j = 1, \dots, p$ choose a range of values b_j (e.g. in a grid)

Fix $\beta_j = b_j$ and re-maximize the likelihood over the other parameters

$$l_p(b_j) = \max_{\beta_{-j}} l(\beta_j = b_j, \beta_{-j})$$


Profile likelihood for β_j

Parameter values $\{b_j : 2(l(\hat{\beta}) - l(\beta_j = b_j, \hat{\beta}_{-j})) \leq \chi_{1,1-\alpha}^2\}$ are $100(1 - \alpha)\%$ - CI for profile likelihood

Geometrically, you are slicing the log-likelihood surface along the β_j -axis and keeping all the points that are within a fixed drop from the peak

Results and scales

At minimum, when presenting the results of GLMs we give the estimated regression coefficients and the confidence intervals

Est	$100(1 - \alpha)\% - \text{CI}$
$\hat{\beta}_j$	$(\hat{\beta}_j^{(L)}, \hat{\beta}_j^{(U)})$

Response is on the link-scale (e.g. logistic regression in log-odds)

Often better to report on the response scale (odds multiply by $\exp(\hat{\beta}_j)$)

Est	$100(1 - \alpha)\% - \text{CI}$
$\exp(\hat{\beta}_j)$	$(\exp(\hat{\beta}_j^{(L)}), \exp(\hat{\beta}_j^{(U)}))$

End of Lecture 5